MD. SHAH NAWAZ ROMEL

Website | Gmail | Google Scholar | LinkedIn | ResearchGate

EDUCATION

Khulna University of Engineering and Technology (KUET) **Khulna, Bangladesh**

Bachelor of Science (BSc) in Materials Science and Engineering **Jan 2020 – Oct 2025**

CGPA: 3.15/4.00

RESEARCH INTERESTS

- Computational Materials Science (Quantum & Electronic Structure Modeling, Atomistic & Molecular Simulations, Materials Discovery & Design)
- Perovskite & Next-Generation Solar Cell Design (Organic-Inorganic, Halide, Double & Anti-Perovskites, Defect/Interface Engineering, Device Simulation & Stability Optimization, ML)
- Energy Storage Materials & Computational Battery Modeling (Electrochemical Interfaces, Ionic Diffusion, Solid-State Electrolytes & Hydrogen Storage Systems)
- 2D Materials & Optoelectronics (MXenes, Graphene, van der Waals Heterostructures, Electronic/Optical Property Engineering & Nanoelectronic Devices)
- Advanced Structural, Electronic & Extreme-Environment Materials (Additive Manufacturing, Electrochemical Degradation, Thin-Film Semiconductor Engineering, High-Temperature Alloys & Ceramics)

RESEARCH EXPERIENCE

Undergraduate Research Work, CMS LAB, KUET **Jan 2024-Oct 2025**

- Studying the physical characteristics of perovskite, anti-perovskite, double perovskite, and 2D MXene materials through Density Functional Theory (DFT) to understand their potential in energy and optoelectronic applications
- Optimizing perovskite and hybrid organic–inorganic solar cell structures by integrating DFT analysis with COMSOL Multiphysics and SCAPS-1D simulations to evaluate their photovoltaic performance
- Utilizing LAMMPS for molecular dynamics simulations to investigate atomic-level interactions and evaluate structural, thermal, and mechanical properties of advanced materials
- Computational Modeling of Battery Systems using COMSOL Multiphysics for electrochemical transport, ion diffusion, thermal management, and solid-state battery analysis.

JOURNAL PUBLICATIONS

[1] **M. S. Nawaz Romel** et al., “Theoretical and device-level insights into $FASnBr_3$ perovskite solar cells: a synergistic approach Combining density functional theory and SCAPS-1D,” Results Phys., p. 108593, Feb. 2026, doi: 10.1016/j.rinp.2026.108593

[2] S. Shahrear, **M. S. N. Romel**, I. A. Ovi, M. M. Hasan, F. Rubaiya, and M. S. Islam, “Understanding

Pressure-Induced Photovoltaic Performance of Lead-Free Ca_3BiI_3 Perovskite: A DFT and SCAPS-1D Simulations Study," Nano Sel., vol. 7, no. 5, p. e70145, 2026, doi: 10.1002/nano.70145.

[3] R. N. Mondal, F.-T.- Zahra, Md. B. H. Parosh, and **Md. S. N. Romel**, "Unveiling Thermal, Phonon, and Optoelectronic Characteristics of Cs_3OX (X = Cl, Br, or I) Antiperovskites for Sustainable Energy Applications: A DFT Exploration," Int. J. Energy Res., vol. 2026, no. 1, p. 5561106, 2026, doi: 10.1155/er/5561106.

[4] **M. S. Romel**, F. Zohra, A. Rashid, and I. Ovi, "First-principle calculation of the physical properties of tri-halide perovskite $FrSrX_3$ (X = F, Cl, Br, and I): A DFT study," AIP Adv., vol. 15, Oct. 2025, doi: 10.1063/5.0281327.

[5] Fatema Tuz Zohra, **Md Shah Nawaz Romel**, Ahnaf Al Rashid, Walid Ahammed, Istiak Ahmed Ovi, "Exploring Pressure-Driven Properties of $FrSrX_3$ (X = Cl, Br and I) Compounds for Advanced Applications in Photovoltaics and Optoelectronics" (Accepted)

CONFERENCE PRESENTATIONS

[1] **M. S. Romel** and F. T. Zahra, "DFT Investigation of Cs_2NaPBr_6: A Promising Double Perovskite for Sustainable Energy Applications," 8th International Conference on Engineering Research, Innovation and Education (ICERIE), SUST, Sylhet, Bangladesh, 2025.

STANDARDIZED TEST SCORES

International English Language Testing System (IELTS) – 4th December, 2025

Overall Band	Listening	Reading	Writing	Speaking
6.5	8.0	6.5	6.0	6.0

TECHNICAL SKILLS

- **Simulation Software**: CASTEP, Quantum Espresso, LAMMPS, SCAPS-1D and COMSOL Multiphysics
- **Machine Learning Tools**: scikit-learn, XGBoost, SHAP, Optuna
- **Crystal Structure Visualization**: VESTA (Expert), XCrySDen (Basic), OVITO (Basic)
- **Data Analysis & Post-Processing**: Origin Pro, Excel, CES EduPack, and XMerge
- **Design & CAD Software**: AutoCAD (Expert), SOLIDWORKS (Basic)
- **Operating Systems**: Windows OS (Expert), Linux/Ubuntu (Intermediate)
- **Programming & Scripting**: Python (Basic), C, C++.
- **Productivity & Office Tools**: MS Office Suite (Expert), Libra Office, LaTeX (Intermediate)

INDUSTRIAL EXPERIENCE

Bangladesh Steel Re-Rolling Mills Limited (BSRM) — Chittagong, Bangladesh
Industrial Attachment — November, 2024
Safety Measures, Scrap Yard, Induction Furnace, Ladle Furnace, Continuous Casting Machine (CCM), Quality Control Management (QCM)

ACADEMIC AWARDS & HONORS

- Technical Scholarship (2x) 2021-2023
  Awarded by Khulna University of Engineering & Technology for merit-based position.
- Board Scholarship 2020-2024
  Awarded by Board of Intermediate and Secondary Education, Dhaka

LEADERSHIP & CO-CURRICULAR ACTIVITIES

- Greater Barisal Association (GBA) 2023-2025
  Vice President (VP)
- MSE 2K19 2021-2022
  Class Representative (CR)

PROFESSIONAL SOFT SKILLS

Critical Thinking, Creativity and Innovation, Strategic Thinking and Teamwork, Leadership and Mutual Respect, Public Speaking, Problem Solving, Determination, Negotiation, Empathy, Time Management, Adaptability and Flexibility, Communication, Work Ethics and Organization.

REFERENCES

- Dr. Md. Shahidul Islam

  Professor, Department of Mechanical Engineering (ME), Khulna University of Engineering & Technology (KUET), Khulna, Bangladesh
  shahidulbitk@me.kuet.ac.bd

- Fatema-Tuz-Zahra (Thesis supervisor)

  Assistant Professor, Department of Materials Science and Engineering (MSE), Khulna University of Engineering & Technology (KUET), Khulna, Bangladesh
  zahra@mse.kuet.ac.bd

- Md. Salman Haque (Course Teacher)

  Assistant Professor, Department of Materials Science and Engineering (MSE), Khulna University of Engineering & Technology (KUET), Khulna, Bangladesh
  salmanhaque@mse.kuet.ac.bd